

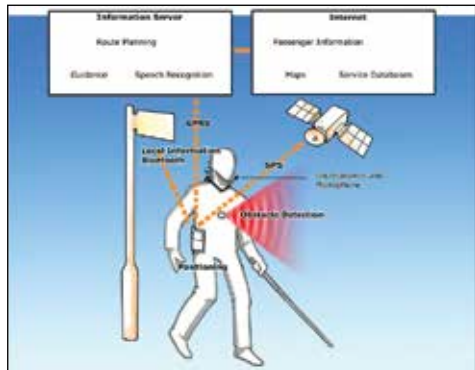
## Any solution?

Definitely. This is where the role of advanced hardware comes into play. These advanced hardware devices have far greater processing capabilities than existing ones. But before we talk about that, let's see if there are any solutions to the problem of navigation for the blind.

- ◆ There are algorithms that can detect how far an object is from you, by using images from a camera.
- ◆ Since you're using images, you can separately detect cars and other objects, to warn users about any possible vehicle directly in front, so that he can take necessary precautions.
- ◆ It's also possible to detect road signals to inform the person when he should cross the road.
- ◆ Using images, the system can detect the text on signboards, for instance, by running images through an optical character recognition system. So apart from being just a navigation system, the system also acts as an assistant and artificial eye.

Though this solution is great, it definitely has limitations:

- ◆ It can better detect objects in front than at the side. So the user needs to move to detect objects on either side.



Advanced navigation for the blind

- ◆ Fast moving vehicles are still a problem in areas without signals.

Let's bring the Internet of Things capability of these advanced devices into the picture.

- ◆ Imagine a completely connected world that might exist in the future.
- ◆ All vehicles in this hypothetical future have GPS showing their current position.
- ◆ The device that the blind person possesses also has a GPS system that detects his current location.
- ◆ The GPS device helps the blind person navigate with appropriate guiding directions.